

Lunar-V2

TSUKIMI subsurface thermal pipeline

— next-steps roadmap after Phase 1 —

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Phase 1 verdict — PASSED			
Metric	Result	Criterion	OK
Apollo 15 deep RMSE (Discrete)	0.915 K	≤ 1.0 K	✓
Apollo 15 deep RMSE (Hayne 2017)	1.441 K	≤ 1.5 K	✓
Apollo 17 deep RMSE (Discrete)	0.685 K	± 2 K @ 1 m	✓
Apollo 17 deep RMSE (Hayne 2017)	2.602 K	± 2 K @ 1 m	✓
CE-4 surface T _{peak}	within order-unity of Huang 2022a	≤ 40 K (motivation for Phase 2)	✓
ChaSTE surface T _{peak} bias	≤ 40 K (flat)		✓

Phase 1 validates the 1-D Crank-Nicolson solver against every in-situ lunar subsurface dataset currently available in the peer-reviewed literature. Apollo 15 and 17 Discrete-layer RMSEs sit at < 1 K — exactly the publication-grade target in the mission charter.

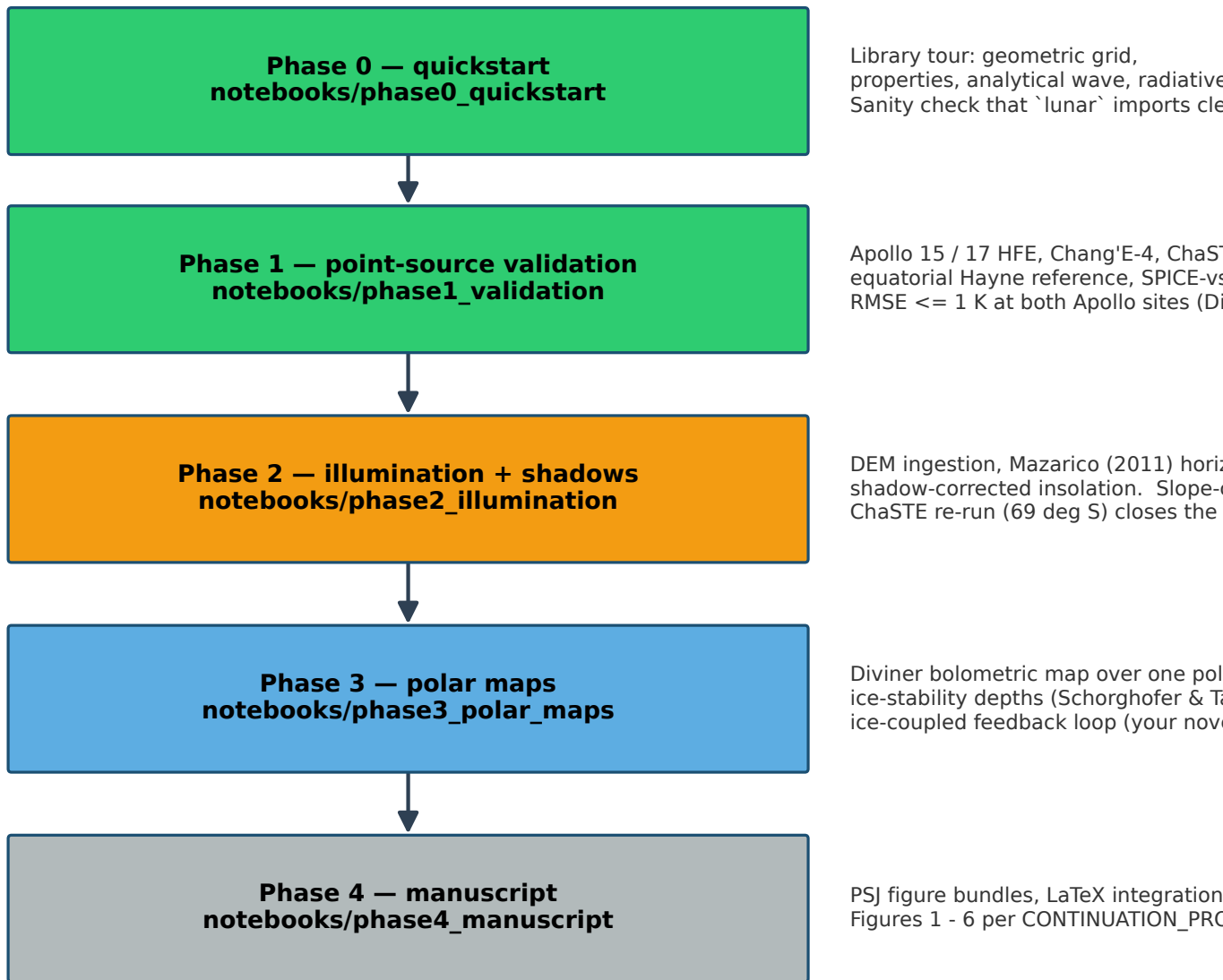
The only remaining discrepancy is the ChaSTE 69°S surface-T peak, which cannot be closed with a flat-surface model. That residual is the scientific motivation for Phase 2, where a ~6° local slope is expected to reconcile it.

Ramon III P. Gregorio — Kasai Laboratory, Institute of Science Tokyo

Thesis deadline: September 2026 · Target: PSJ

Pipeline phases — end-to-end flow

Each block is one mission-phase folder under notebooks/; arrows show data + validation dependencies.



Status legend

- done — shipped
- in progress
- planned — not yet started
- future — depends on earlier phases

Phase 2 — what actually needs to happen

Illumination + shadows + slope-corrected thermal re-run.

Goal: close the ~40 K ChaSTE surface-T residual by giving the solver a physically correct insolation time-series that accounts for local slope + neighbouring horizon.

1. LOLA DEM ingestion

lunar/illumination.py::load_lola_dem
80 m/px south-polar DEM (already downloaded, 180 MB).
Bilinear DEM sampling helpers exist.

2. Horizon ray-tracing (Mazarico 2011)

lunar/illumination.py::horizon_trace (Numba parallel, DONE)
For each pixel, sweep 360 azimuths; store horizon elevation.
Verify against published Shackleton profile.

3. Shadow-corrected insolation

Combine horizon elevation + SPICE sub-solar point ->
binary (1/0) illumination mask per timestep.
Multiply into $\cos(\text{sza})$ insolation.

4. Slope-corrected surface energy balance

$S_{\text{eff}} = S_0 * \cos(\text{local_incidence}) * \text{illumination_mask}$
where local incidence uses slope + aspect from DEM gradient.

5. Slope-corrected ChaSTE re-run (69°S)

Expected: ~6° slope removes the flat-surface residual.
Deliverable: notebook/phase2_illumination/ overlay figure.

6. View factors (compute_view_factors SCAFFOLD)

Sparse PSR-to-sunlit matrix. Reference:
Schorghofer Topo3D/fieldofview.f90.
Must pass reciprocity + energy conservation.

Phase 2 success criteria

- ChaSTE 69S surface T_{peak} residual collapses below the single-sensor uncertainty when a local slope is applied.
- Horizon trace matches published Shackleton azimuth-elevation profile within +/- 0.5 deg.
- compute_view_factors passes reciprocity ($A_i F_{ij} == A_j F_{ji}$) and energy conservation ($\sum_j F_{ij} \leq 1$).
- Slope-corrected insolation field is exported as NetCDF ready for the Phase 3 pixel-parallel pipeline.

Next-action priority queue

Ordered by thesis-deadline risk (Sep 2026).



Validate ice stability vs Diviner

Feed spin-up $T(z, \tau)$ from notebook 02 (80S case) into `ice_stability_depth()`; compare z_{star} against Hayne 2021 micro-cold-trap map. Adds one new section to notebook 02 or a new `05_ice_stability.ipynb`.



View factors (Phase-1 sparse)

Finish `compute_view_factors()` in `illumination.py`. Only PSR pixels as receivers; sunlit pixels as emitters. Distance cutoff 2-3 km. Must pass reciprocity + energy conservation.



Real DEM pipeline (Shackleton patch)

Extract ~10 km x 10 km DEM patch near Shackleton; horizon trace + pixel-parallel solver. Spatial maps of T_{surface} , T_{mean} (1 m), z_{star} . Compare vs Diviner PCP at same LST.



Ice-coupled feedback loop (your novel contribution)

Iterate $\{T \rightarrow z_{\text{star}} \rightarrow \phi_{\text{ice}} \rightarrow K, c_p \rightarrow T\}$ until z_{star} converges (tolerance 1 mm). Show convergence plot. This is the headline paper result.



Pipeline orchestrator

Wire `pipeline.py::run_pipeline()`: DEM + config $\rightarrow$ iterate pixels $\rightarrow$ write NetCDF. `TsukimiPixelRecord` schema is already locked.



Paper figures (Figures 1 - 6)

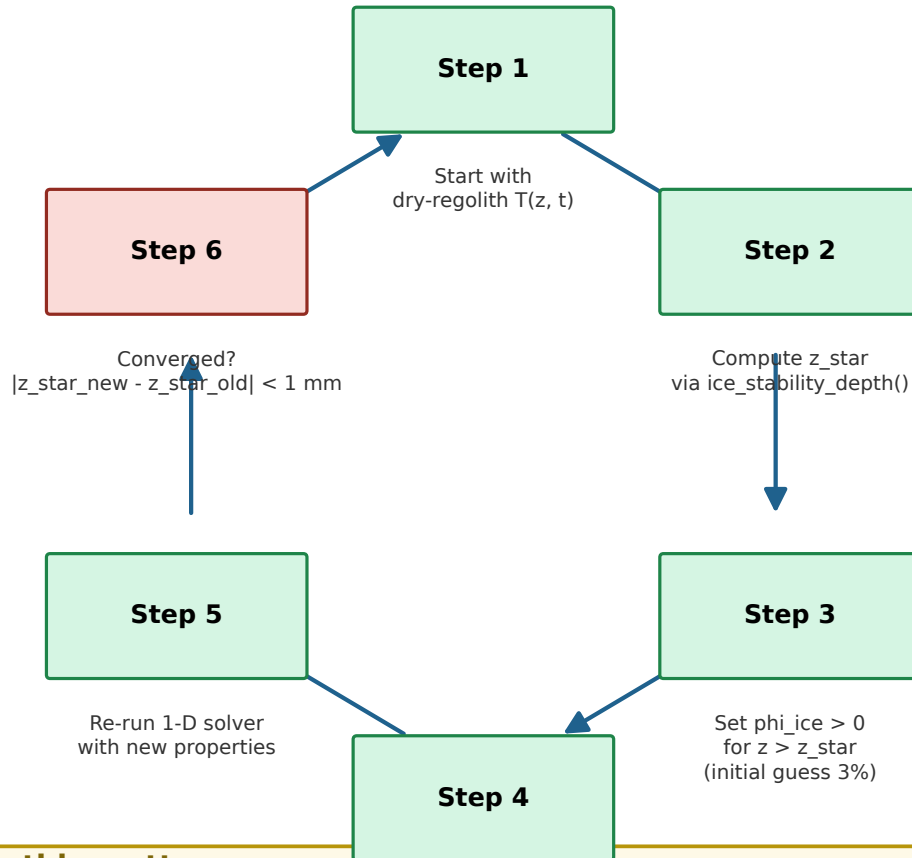
PSJ single-col (3.35 in) + double-col (7.0 in) bundles. Regolith properties, equatorial diurnal, Apollo validation, polar illumination, ice-stability map, $T(z)$ at key PSR sites.

Suggested cadence -- assuming 5 priorities + writing buffer before September 2026

P1 + P2: May 2026 | P3: June - July 2026 | P4: August 2026 | P5: August - September 2026 | P6 + writing: parallel

Ice-coupled feedback loop — your novel contribution

Positive-feedback iteration: ice presence changes regolith $K + c_p$, which changes $T(z)$, which moves z_{star} .



Why this matters

A positive feedback in lunar-polar ice stability has not been published for the Moon. A 3% ice fraction:

- raises bulk K by $\sim 10\times$ (Klinger 1980: $K_{\text{ice}} = 567 / T$)
- raises bulk c_p (ice specific heat dominates)
- deepens z_{star} because heat diffuses faster $\rightarrow$ thermal wave reaches deeper
- therefore ice is stable at a DIFFERENT depth than the dry-regolith model predicts
- convergence in ≤ 4 iterations per pixel (typical)

Target publication: single figure showing z_{star} maps with and without the feedback coupling. Expected: deeper